

HW07

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Exercise 2

- Points attempted: 5
 - Self-assessed score: 5/5.
 - Time spent: 60 minutes.
 - Difficulty: 6/10.
 - Questions/Feedback: Clarity, correctness.
1. [2] Read Tai-Danae Bradley's blog post What's a Quotient Group, Really? Part 1. (Professor Bradley wrote this blog when she was in graduate school, and her blog was a part of the motivation for doing a blog post as the term paper in this class.)

Done.

2. HW7, Exercise 2: i added the defintion of an abelian group.
3. Do at least *three* of the following for your blog:

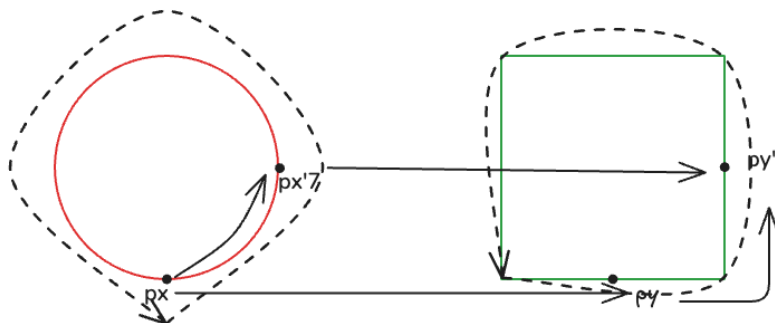
- a Write an outline of what your blog post is going to be about.

Although the full prompt that I will be following is "The Fundamental Group of a Topological Space", I want to maybe open up the scope just a tad, and include more details about algebraic topology itself. Thus, I want to include a section on specifically the foundations of algebraic topology. Then, as a we build solid foundation, the blog should transition into a more focused discussion about how groups show up in algebraic topology, specifically as the fundamental group of a topological space.

Lastly, I want to also get into some applications. As I am currently deeply interested in machine learning, and the mathematical foundations of it, I want to explore some applications of topology in machine learning. There is also a concept in deep learning known as the **manifold hypothesis** which states that 'most' high dimensional data can actually be summarized as a manifold with a lower 'intrinsic dimensionality.'

- b Write two paragraphs explaining a core concept or idea.

The bedrock of topology is a concept known as **homotopy**. Although topology is often introduced in terms of donuts and mugs, I believe that a slightly more 'formal' defintion is worth investing into before we continue. Without too much detail[TK], suppose we have two topological spaces X and Y . Now, consider two maps from $f, f': X \rightarrow Y$. We consider f and f' **homotopic** if we can "continuously deform" one into the other. This may be hard to visualize, but imagine that we have two topological spaces, one that is the perimeter of a circle, and another that is the perimeter of a square, (we will see later that these two are inherently the same space, but for now, we can treat them as different spaces). Notice that all mappings from a circle to a square can be deformed into each other, if they have a specific quality: that they go around the object the same amount of times.



Imagine a mapping where we go around the circle once, but around the square twice. [TK] No matter how we stretch this mapping, we can never get something that maps from the circle to the square that go around the same number of times. The formal name for this concept is a **winding number**. Thus, all mappings in this setting are homotopic if they share a winding number. Further more, we can group the mappings that are homotopic [TK, need to explain why homotopy is an equivalence relation] into something known as **homotopy classes**.

- c Include an illustration, or plan how you will make your illustration.

A made a rough illustration of the example above.